# INFORMATION SEARCH AND ANALYSIS SKILLS (ISAS)

## "MACINTOSH 14.6.1 SONOMA"

Group 4


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# PREFACE

First of all, we give thanks to Allah SWT, who has given us the opportunity to complete this task. Blessing and salutation may also be upon the most honorable prophet Muhammad SAW who has given us life guidance from our birth to the end of life

The title of this paper is “MACINTOSH 14.6.1 SONOMA” and we submit it as ISAS task requirements. Our purpose in creating this report is to get to know the operating system, especially the MacOS 14.6.1 sonoma operating system. We made the entire report carefully and through full analysis

Depok, 27th August 2024

Authors

# TABLE OF CONTENTS

preface ..... **ii**
Table of contents ..... **iii**
Table of figures ..... **iv**
**CHAPTER I INTRODUCTION ..... 1**
I.1 Background ..... 1
I.2 Writing Objectives ..... 1
I.3 Problem Domain ..... 1
I.4 Writing Methodology ..... 1
I.5 Writing Framework ..... 2
**CHAPTER II BASIC THEORY ..... 3**
II.1 Definition of Operating System ..... 3
II.2 Function of Operating System ..... 3
II.3 Various Type of OS ..... 4
II.4 Definition of MacOS ..... 6
II.5 MacOS Sonoma Version ..... 7
**CHAPTER III PROBLEM ANALYSIS ..... 8**
III.1 MacOS Sonoma Features ..... 8
III.2 MacOS Sonoma Updates and Patches ..... 12
III.3 MacOS Sonoma advantages and disadvantages ..... 13
III.4 MacOS Comparison ..... 14
**CHAPTER IV CONCLUSION AND SUGGESTION ..... 16**
IV.1 Conclusion ..... 16
IV.2 Suggestion ..... 16
**BIBLIOGRAPHY ..... 17**

# TABLE OF FIGURES

Figure 2.1 MacOS Sonoma……6
Figure 3.1 Widget Features of Apple…….8
Figure 3.2 MacOS Updates and Patches…….12
Figure 3.3 Sonoma vs Ventura……14

# CHAPTER I

# INTRODUCTION

## I.1 Background

Technology has advanced quickly throughout history. In particular, computer technology. As a result, we will discuss the operating system in this article, with a particular focus on MacOS. As students of information technology, we have a duty to be aware of the constantly evolving digital world where we live.

MacOS is the second-largest operating system globally. This operating system is extremely unique and is only under an Apple license. This OS was created only for Apple products and can only be used by them due to its exclusivity. This operating system is exclusive to Mac, iMac, and MacBook models. That's what keeps the MacOS market in demand and growing

## I.2 Writing Objectives

The purpose of this retrospective is to look at the features of MacOS 14.6.1 from an objective point of view to appreciate and measure how far we have come and how far we have regressed in terms of computer GUI (Graphical User Interface).

## I.3 Problem Domain

The issues author discussed in this ISAS are as follow:

1. Definition of an Operating System
2. Function of an Operating System
3. Definition of the MacOS
4. The characteristics of the MacOS Sonoma
5. The History of the MacOS Sonoma
6. The perks and weaknesses of the MacOS Sonoma

## I.4 Writing Methodology

Our writing method is descriptive analysis. Data research techniques such as descriptive analysis helps in the description, illustration, or summarization of information points, allowing the development of patterns that fulfill all the requirements of the data. through online resources and user feedback

## I.5 Writing Framework

The writing framework for this paper will consist of four chapters, starting with the Introduction of this research paper. And then we'll dive into the basic theory, and then we'll explain about further details of the subject. And then finishing off with conclusion and suggestion.

**Chapter I: Introduction**

This chapter contains background, writing objectives, problem domain, writing methodology, and writing framework

**Chapter II: Basic Theory**

This chapter explains the definition of operating system, function of operating system, Various type of OS, definition of MacOS, MacOS version

**Chapter III: Problem Analysis**

This chapter explains the development of the MacOS Sonoma and what the Perks and Weakness of Updating the MacOS Sonoma version

**Chapter IV: Conclusion and Suggestion**

This chapter contains conclusions and suggestions regarding the research we are conducting and observe

# CHAPTER II

# BASIC THEORY

## II.1 Definition of Operating System

An operating system is an application of software that programs, controls, and facilitates communication between users and hardware by managing hardware and other fundamental system functions.

The main task an operating system carries out is the allocation of resources and services, such as the allocation of memory, devices, processors, and information. The operating system also includes programs to manage these resources, such as a traffic controller, a scheduler, a memory management module, I/O programs, and a file system. The operating system simply provides an environment within which other programs can do useful work.

The operating system enables the control of computer programs through the management of processing time, error checking, input and output control, computing, complicating, storing, and processing data, as well as the execution of various software forms that regulate the execution of Related services. The operating system is often the initial software layer that loads into your computer's memory upon startup. The operating system performs the primary functions of the software, while other programs execute after it.

Operating systems are frequently utilized as a channel for information between the user and the hardware of the computers. An operating system handles responsibility as an intermediary between the end-user and the system hardware. Examples include managing computer resources, offering services, and carrying out user programs; presenting an interface to the user; concealing the intricacy of software; enabling several modes of execution; and keeping an eye on user program execution to avoid errors.[1]

## II.2 Function of Operating System

The function of an operating system is to serve as an interface or communication link between computer hardware and users. An operating system's main function is to give users a convenient and effective platform to run programs on. A piece of software that controls how computer hardware gets assigned is called an operating system. To guarantee that the computer system operates correctly and to stop user programs from interfering with the system's proper operation, the hardware must be coordinated appropriately. Optimizing resource usage is the secondary objective of the operating system, which is to make the computer environment more user-friendly.[1]

Here are some Operating System usages:

1. **Security**

The operating system uses password protection and other relevant safeguards to secure user data. Programs and user data are also shielded against unauthorized access.

2. **Control over System Efficiency**

To maximize performance, the operating system keeps an eye on the system's general health. Keep note of the intervals between system responses and service requests to obtain a comprehensive understanding of the system's health. By offering crucial information for diagnosing problems, this helps improve performance.

3. **Accounting Jobs**

The operating system keeps track of how much time and resources are spent by various users and tasks; this information can be used to gauge how well a particular user or set of users is using their resources.

4. **Aids Error Detection**

The operating system (OS) continuously scans the system to find errors and keep a computer system from malfunctioning.

5. **Coordinating with Other Software and Users**

Operating systems also arrange and allocate other software, such as interpreters, compilers, and assemblers, to computer users.

6. **Memory Control**

The primary memory, also referred to as the main memory, is managed by the operating system. There are many different words or bytes in the main memory, and each one has an address. Fast storage that the CPU can access immediately is the main memory. Before a program can be executed, it needs to be loaded into the main memory.[2]

### II.3 Various Type of OS

Digitized and modern era has been growing rapidly every year. Operating systems have evolved into many different kinds, particularly in the field of information technology. Each technology company invents and advances these technologies with their own particular characteristics and benefits. Here are some Operating System from various company:

1. **Windows** by Microsoft. Since the early 1990s, Windows, the most popular operating system worldwide, has led the desktop OS market. According to recent figures, Windows 10 holds a massive 71.29% of the market share, making it the most popular operating system. Windows 11 comes in second place with 15.44% of the market. Windows is

distinguished by its graphical user interface, compatibility, practicability, and simplicity in use. Of all the OSs, it also provides the widest collection of applications. Windows 7, Windows Vista, MSDOS, Windows Server, and Windows XP are just a few of the many variations.

2. **Mac operating system, or macOS.** The operating system that runs iMacs and MacBooks is called Apple macOS, and it has a very beautiful design, strong security, excellent performance, and easy usability. It's also thought to be among the fastest and most reliable.

3. **Linux**. Linux is a free, open-source UNIX-based software that is extensively used as a development environment by computer science professionals. It is perfect for data centers, cloud-based services, and websites and high-volume critical applications that depend on massive servers.

4. **Chrome OS**. Google created ChromeOS, an easy-to-use, safe, and straightforward operating system, specifically for tablets and netbooks.

5. **Android**. The most popular smartphone operating system worldwide is Google's Android OS. This open-source software, which is a version of the Linux kernel, is mainly used for touchscreen devices, including mobile phones, smartwatches, smart TVs, and any other kind of smart display. Because of Android's rich customization options and eye-catching design, a lot of consumers go with it.

6. **IOS**, the second most popular mobile operating system, was created exclusively to be used with Apple hardware, including the iPhone, iPad, iPod, and other portable devices.[3]

**II.4 Definition of MacOS**

**Figure 2.1 MacOS Sonoma**
**(Ref: https://www.trustedreviews.com/news/macos-sonoma-4333722)**

What is MacOS? The operating system (OS) for Apple laptops and desktop computers is called MacOS. Every Mac is powered by a proprietary graphical operating system (OS).

Operating systems communicate with a computer's hardware by assigning it the resources it needs to finish tasks that are assigned to it, like running an application. Operating systems assign memory, computing power, and file storage among other resources.

MacOS is generally fast and responsive since the operating system has been optimized to the specific hardware. Every year, Apple releases an update for macOS.

In 1984, MacOS was released to operate Macintosh personal computers (PCs). The operating system (OS) contributed to the rise in popularity of graphical user interfaces (GUIs), even motivating Microsoft to switch from its user input technique to GUIs for Windows Operating Systems (OS). [4]

**II.5 MacOS Sonoma Version**

. As of this writing, the most recent Macintosh operating system to be fully launched is MacOS Sonoma. The upgrade was published on September 26 2023 and was unveiled at June 5th 2023 WWDC 2023. Named after Sonoma, California's wine region, Apple's version is the follow-up to macOS Ventura. Along with a few Mac-specific improvements over macOS Ventura, Apple has integrated several of the features from iOS 17.[4]

# CHAPTER III

# PROBLEM ANALYSIS

## III.1 MacOS Sonoma Features

The most recent MacOS Update, Sonoma, incorporates multiple features from iOS 17 and adds a plethora of new capabilities while updating some of the older MacOS Ventura features.The main attributes and improvements consist of:

**Figure 3.1 Widget Features of Apple**

### 1.Screensavers

Some new screen savers were introduced by Sonoma. Gorgeous screensavers with slow-motion footage of many locations across the globe, such as the nighttime skyline of Dubai, the backcountry of Yosemite National Park, and Sonoma. You have the option to shuffle the themes of Landscape, Earth, Underwater, or Cityscape.

plus, a fresh login procedure. A streamlined login process creates room to highlight your wallpaper or screensaver, complete with the date and time. Multiple account avatars are displayed using a new selector. New screen saver images blend in seamlessly with your desktop when you log in

**2. Widgets**

Widgets were also included in Sonoma for the Mac desktop. While working, customize your desktop to your liking and quickly access information. When a widget detects an empty space close to your cursor, it will automatically relocate there and group with other widgets to form an attractive desktop layout.

Moreover, widgets adjust to your desktop. In order to preserve legibility and the aesthetics of your desktop, widgets intelligently shade based on the color behind them. You can quickly add widgets to files. When you arrange widgets next to your current files, the desktop will automatically adjust to accommodate them. Stacks and files encircle widgets, rearranging them to fit in the available area; these are just a few instances of the numerous ways you may use the new widgets feature that MacOS Sonoma has to offer.

**3. Video conferencing**

Presenter Overlay. Remain a part of the conversation when you're sharing your screen on a video call by placing yourself on top of your presentation. Use a small overlay to place yourself in a moveable bubble, or use a large overlay to keep yourself prominent while your shared screen is framed beautifully next to you.

Reactions and gestures. Layer screen effects like balloons, confetti, stormy rain, fireworks, or laser beams directly into your camera feed. Trigger them with a click, or go hands-free and trigger reactions using your gestures alone.

Screen share preview. You'll always know exactly what is being shared with a preview of what you're screen sharing in the Video menu bar item.

**4. Safari**

Apple added profiles to Safari as well. to arrange your surfing into categories such as personal and professional. On Mac, you may store your favorite websites to the Dock for convenient access. Each profile also has its own history, cookies, extensions, Tab Groups, favorites, and Web apps. For easier browsing, access them in a condensed manner. Additionally, Safari's Streamlined Search feature makes searches more responsive than before and presents suggestions that are both more relevant and easier to understand. as well as the choice of several tabs. To move tabs around the tab bar or into a Tab Group, select multiple tabs at once.

Additionally, Apple added Locked Private Browsing to the private browser upgrade. To prevent unauthorized users from accessing your tabs, Private Browsing locks when you're not using it. Use your device's passcode, Face ID, or Touch ID to unlock. In Private Browsing, advanced tracker blocking and fingerprint protection are available. protects against fingerprinting

and more aggressive tracker blocking to keep your most sensitive browsing private. In private browsing, use a search engine. For private browsing, choose a special search engine.

**5. Passwords**

Apple introduced passkey and password sharing. Give your closest friends and family access to your passwords. Make a group to share information with, and change or add passwords at any moment. Instantaneous verification code autofill via Mail. You may easily log in without leaving the browser thanks to Safari's autofill feature for one-time verification codes that you receive in Mail. Online Apple ID passkeys. If you're using your Apple ID to log in online, use a passkey.

**6. Messages**

To messages, Apple added stickers. By controlling clicking on a message bubble or using the plus button, you may share any Live Sticker, Memoji, or Animoji. as well as search parameters. To locate the exact message you're looking for, combine search filters (people, phrases, and content kinds like photographs or links) to narrow down your search even further. Catch-up arrow as well. Click the arrow in the upper-right area of the conversation to quickly get to your first unread message in a group chat.

Moreover, swipe to reply was included. Simply swipe right on any bubble to respond to a message inline. and upgrade to audio messages. You can now pause while recording an audio message, pick up where you left off, and record the message again before sending. Play back audio communications you've received at up to two times their original speed or see a transcription.

One-time verification code cleanup is also included. After utilizing one-time verification codes with autofill anywhere on the system, they are automatically removed from the Messages app. Features of iMessage within MMS groups. Additionally, MMS groups can use additional iMessage capabilities, such as edit, responses, effects, tap backs, and more. and iCloud Messages enhancements. Messages settings including SMS filters, send and receive accounts, and text message forwarding will sync between devices if you enable Messages in iCloud.

**7. PDFs**

They included intelligent form detection for PDF. Now, fillable forms and documents can be recognized automatically throughout the system, even in scanned documents, files, and mail. as well as Enhanced AutoFill. Fill out forms with names and addresses even more quickly thanks to strong on-device language models that recognize fillable fields and activate AutoFill.

**8. Notes**

Document scans and Inline PDFs were included for notes. It's simple to see and engage with PDFs and document scans in your note because they're displayed full-width. You can quickly

navigate to a page in a document you're examining and preview numerous PDFs in one note with ease. Notes with links. To connect concepts, information, or any other kind of content, create links to other notes. It's ideal for organizing recipes, research notes, and even lightweight wiki creation for your group. To quickly add links while writing in your message, type ">>."

**9. Keyboard**

predictive text that appears inline. Based on what the keyboard anticipates you will type, get single and multi-word predictions directly in the text field. This allows you to type even faster by simply pushing the Space bar. more precision using Autocorrect. Autocorrect uses a new transformer language model in English, French, and Spanish keyboards to correct errors for you more precisely than before. Furthermore, autocorrect becomes even more accurate across a wider range of languages thanks to improve on-device language models.

easier editing with Autocorrect. As you type, autocorrected words are momentarily underlined to let you know what has changed. Improved word replacements. Sentence autocorrection can fix a wider variety of grammatical errors. In order to make edits and suggestions easier to see and make, when necessary, the keyboard additionally highlights them.

Clear language processing. The keyboard will learn how to use explicit words for each program and add them to your own bank of vocabulary words. Autocorrect and suggestions are based on learning explicit language. Emoji recommendation. To make it simpler to locate what you're looking for, the Character Viewer contains a suggested list that displays emojis that are pertinent to the context of what you're typing.

Quick emoji selection. Pressing the Globe icon on your Mac keyboard pulls up a lightweight emoji picker that shows emoji relevant to the word you just typed. Updated text cursor. The text cursor briefly displays an indicator that shows the current input language when you switch keyboard languages. The indicator can also signal helpful input details like when Caps Lock is on.

Novel text languages with predictive features. Keyboards that are in Hebrew, Polish, or Romanian can use predictive text. novel keyboard configurations. Chuvash, Hmong (Pahawh), Ingush, Kabyle, Mandaic, Mi'kmaw, N'Ko, Osage, Rejang, Sámi, Tamazight (Standard Moroccan), Wancho, and Wolastoqey are among the languages for which new transliteration keyboard layouts are available. With the transliteration feature found on keyboards for Kannada, Malayalam, Tamil, and Telugu, you can type pronunciations using Latin characters that are instantly translated to the desired script as you type.[5]

**10. Gaming**

Additionally, Apple updated the MacOS with Game Mode, which offers faster response times for wireless game controllers and Air Pods and smoother frame rates for gaming. On macOS Sonoma, Game Mode puts game performance first in order to improve in-game responsiveness

and smoother, more consistent frame rates. The OS classifies an application as a game based on its LSApplicationCategoryType. When an application is in game mode, it becomes the front-most program and goes full screen.

### III.2 MacOS Sonoma Updates and Patches

During its lifespan, the MacOS Sonoma has required patches due to various bugs. This section of the research study will outline the history of these patches that the MacOS Sonoma has experienced.

**Figure 3.2 MacOS Updates and Patches**
**(Ref: https://forums.macrumors.com/threads/macos-sonoma-14-6-1-bug-fixes-changes-and-more.2433143/)**

**Sonoma 14.0 on macOS**

With its screen saver and beneficial desktop widgets, macOS Sonoma offers new features that enhance creativity and productivity.

**Sonoma 14.1 on macOS**

Enhancements, bug fixes, and security upgrades for macOS Sonoma 14.1 include features that might be applied to songs, albums, and playlists through the Music app.

**Sonoma 14.2 on macOS**

Enhanced Messages and Weather, including with Auto-Fill for PDFs.

1. pdf, capable of recognizing fields in various forms and PDFs.
2. In a conversation, it's simple to navigate to the first unread message by using the jump arrow on the message.
3. weather, sunrise and sunset timings, air quality, and wind maps, along with forecasts for rainy weather for the next three days.

**Sonoma 14.3 on macOS**

Apple Music has been enhanced in macOS Sonoma 14.3, along with a number of additional enhancements, bug fixes, and security updates for Mac users.

- You may add, rearrange, and remove music from playlists by inviting people to join you using Apple Music's collaboration function.

- AppleCare & Warranty in the Settings displays the level of protection available for all devices that are connected to your Apple ID.

**Sonama 14.4 on macOS**

The podcast episode captions on macOS Sonoma 14.4 can be fully viewed, searched for specific words or phrases, and utilized with accessibility tools like Text Size, Add Contrast, and VoiceOver.

We can receive pre-selected updates through the Messages for Business app, including order progress, flight notifications, fraud alerts, and other transactions from reliable companies.

The Favorites bar in Safari now offers the ability to display simple website icons.

**Sonoma 14.5 on macOS**

Along with bug fixes, macOS Sonoma 14.5 boasts the following enhancements: A brand-new, original word game called Quartiles is now accessible on Apple News+ every day. Access to fresh user information for Quartiles, Crossword, and Mini Crossword is available via the News+ Puzzle scoreboard.

**Sonoma 14.6 on macOS**

With this version, users of a 14-inch MacBook Pro with M3 chip can now use up to two external screens when the notebook is closed. It also includes security upgrades and significant bug improvements.[6]

**III.3 MacOS Sonoma advantages and disadvantages**

Naturally, there are benefits and drawbacks to updating to the newest version of any software; this section of the research report aims to clarify the benefits and drawbacks of MacOS Sonoma.

Advantages:

- Widgets on the desktop
- Safari can use multiple profile
- Locked private browsing tab[7]

Disadvantages:

- There’s no PDF autofill
- Autocorrect feels intrusive with a real keyboard
- Reported incompatibilities with VPN software[7]

### III.4 MacOS Comparison

After reviewing all explanations and benefits, The differences between MacOS Sonoma and MacOS Ventura, the previous version, will be compared in this section.

**Figure 3.3 Sonoma vs Ventura**
**(Ref:https://blog.shopdutyfree.uk/comparison-macos-sonoma-vs-ventura/)**

**1. Widgets**

More flexible and understandable dynamic widgets are introduced by Sonoma. Sonoma's widgets provide real-time updates and interaction, making them more engaging for users than Ventura's static widgets.

**2. Screensavers**

Ventura offered several excellent screensavers, but Sonoma beats that in terms of aesthetics. With more visually beautiful and captivating screensavers, Sonoma elevates the idle times on your Mac.

**3.Video conferencing**

Although Sonoma and Ventura both have powerful video conferencing capabilities, Sonoma has advanced features like noise suppression and background blur that elevate your Zoom calls.

**4. Gaming**

Sonoma advances gaming with more game titles supported and better graphical capabilities. In this area, Ventura trailed behind, while Sonoma offered improved graphics and more fluid gameplay.

**5. Messages**

The Messages app in Sonoma receives an update that includes improved media sharing and inline replies. Though practical, Ventura's version lacks these contemporary additions.

**6. PDFs and Notes**

Sonoma makes taking notes easier with its enhanced organizing and management for PDFs. It differs from Ventura with features like quick annotations and seamless syncing.

**7. Safari**

Faster performance and improved privacy features are included in the Sonoma edition. Sonoma is a great option if you prioritize security and swiftness when online.

**8. Accessibility**

Sonoma accessibility improvements facilitate the use of Macs by people with disabilities. Sonoma has several improvements over Ventura, including voice control, enhanced text-to-speech, and adaptable display settings.

**9. Security**

For any operating system, security is the primary concern. Better encryption and more frequent security upgrades are the extra features that Sonoma adds to Ventura's strong security foundation. Because of this, Sonoma is a better option to safeguard your private information.[8]

# CHAPTER IV

# CONCLUSION AND SUGGESTION

## IV.1 Conclusion

An operating system is an essential component of a device since it is a system that powers it. In the world of operating systems, MacOS is one of many varieties. The operating system MacOS was developed by the Apple corporation, which is well-known for its advanced capabilities and outstanding safety system.

MacOS is constantly evolving with slick features and ideas. Apple is famous for its technology that is always updated every year consistently. Starting from the exterior and interior design. Although they are famous for their expensive products, they always manage to attract the attention of potential users. The most recent and superior version of MacOS is Sonoma, despite numerous releases. Designers should definitely try this macOS.

## IV.2 Suggestion

Our suggestion is, try to explore this operating system more deeply, many media outlets have given reviews of this OS. and this OS has many and exclusive advantages. Many sources of information that can be searched ranging from reviews of people on YouTube, research papers, and even from Apple's own website have provided a lot of information.

For those who are still unsure about which device to buy and what operating system. Think about MacOS as a potential option on your list. because the second-best operating system is MacOS. It's elegant, and the modern look makes it seem simple for consumers to go about their daily activities. Furthermore, a strong security system adds value to this operating system.

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